Fibroblast states and the postnatal CM environment

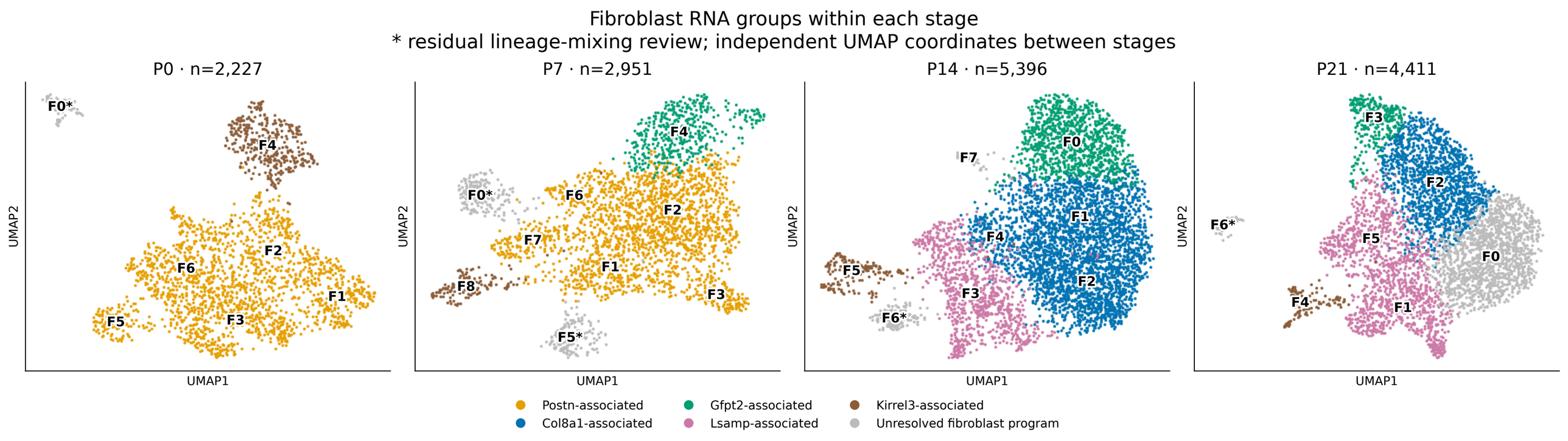
Completed reanalysis: stage-specific RNA discovery, matrix comparisons, and Xenium association tests.

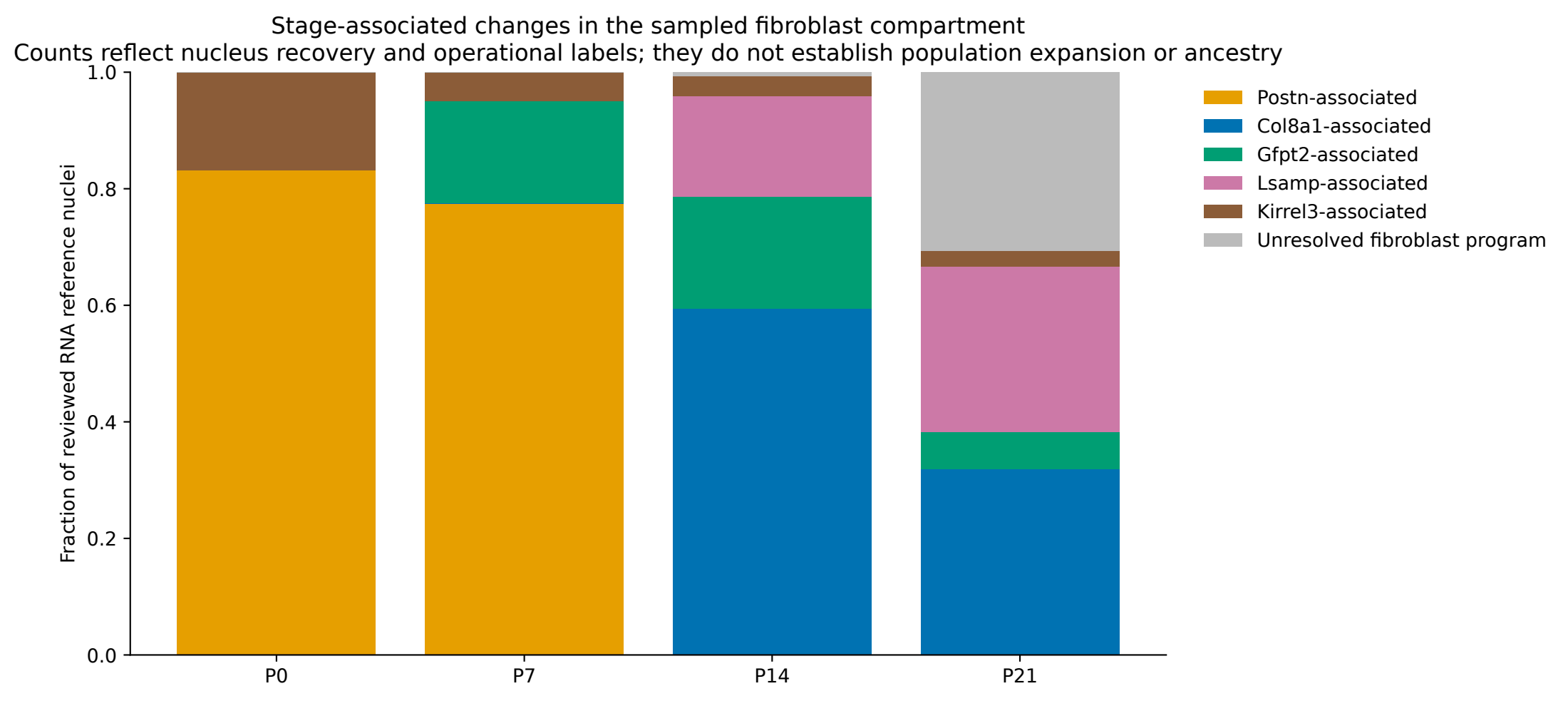
14,985 original QC-retained fibroblast nuclei · eight libraries · four ages
31 computational groups; 657 nuclei in residual lineage-review groups
No nuclei are deleted from the original data. Lineage-review groups are withheld from the fine reference.

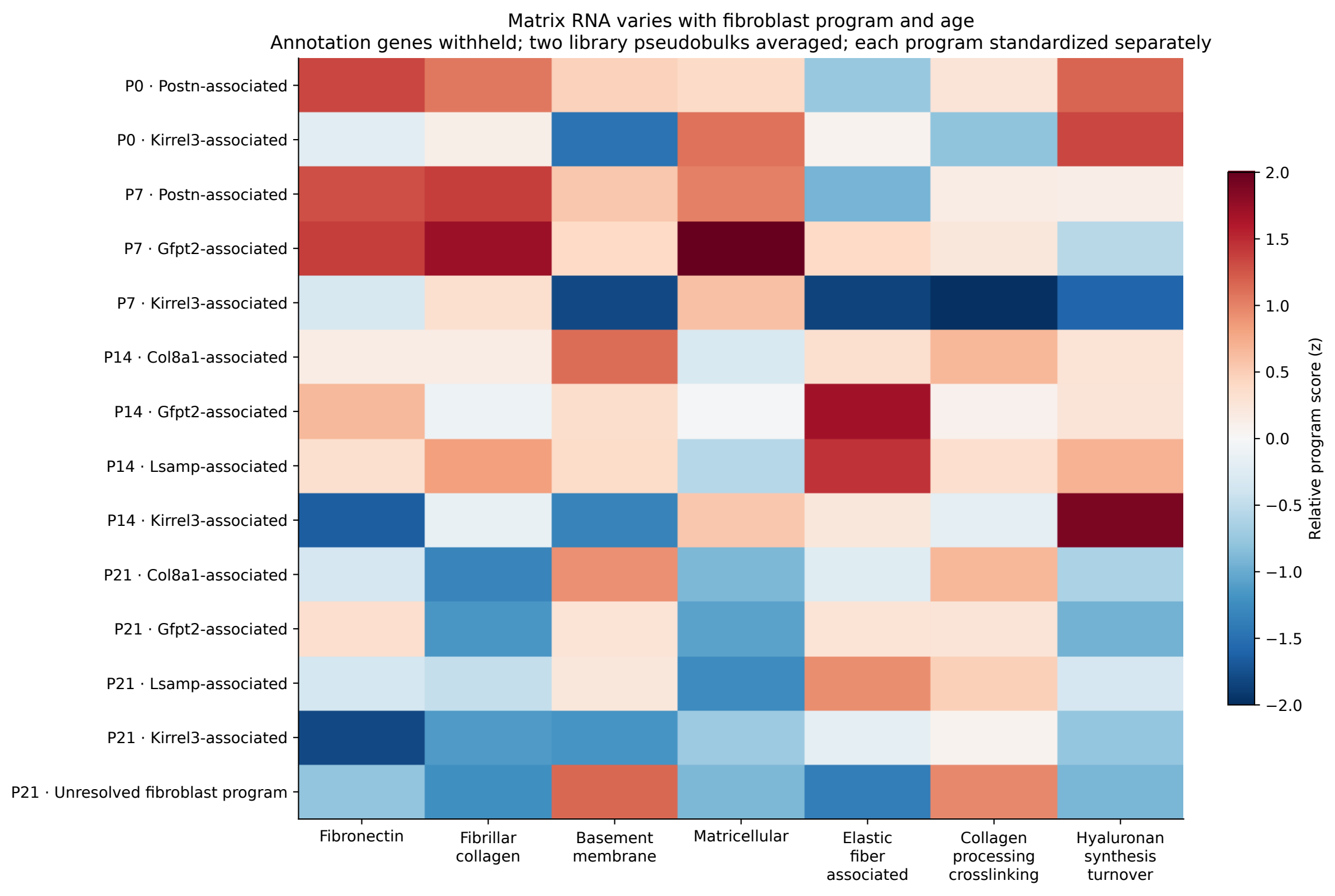
CM maturation uses the exact 21 genes displayed in manuscript Figure 4b.
Primary score: mean log1p(CP10k); fixed matched-control and denominator sensitivities.
This is an independent score using the intended panel, not an exact reproduction of the author index.

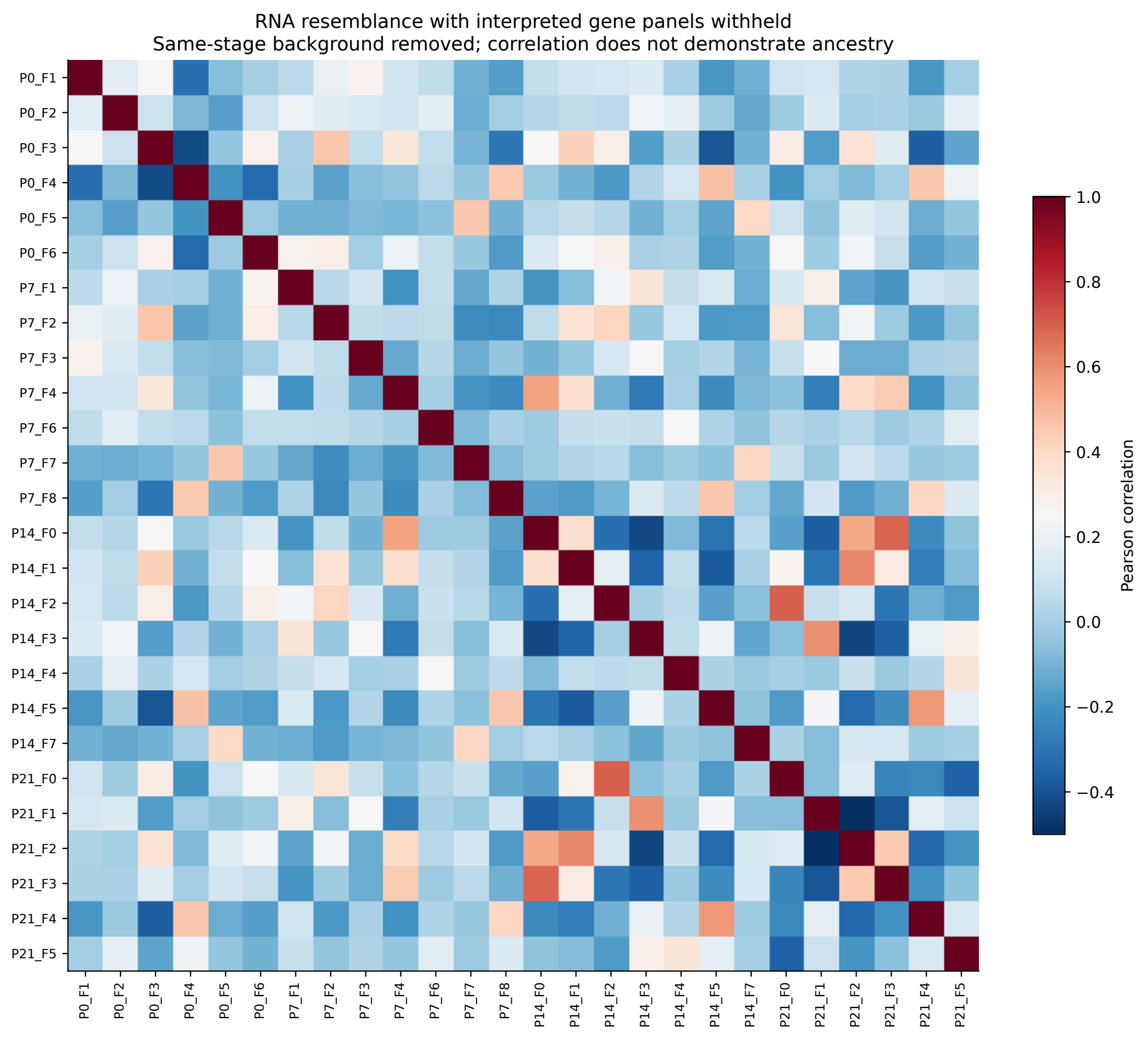
Main findings
- Early Postn-associated groups contain several distinct transcriptional programs.
- A Gfpt2-associated group is already present at P7 and resembles the later group.
- Basement-membrane and other matrix programs differ by state as well as stage.
- Fine individual Xenium labels are much less well supported at P14/P21.
- Fibroblast features add little to spatially held-out CM maturation prediction.
- Modest Ankrd1 associations depend on the exposure/boundary definition.

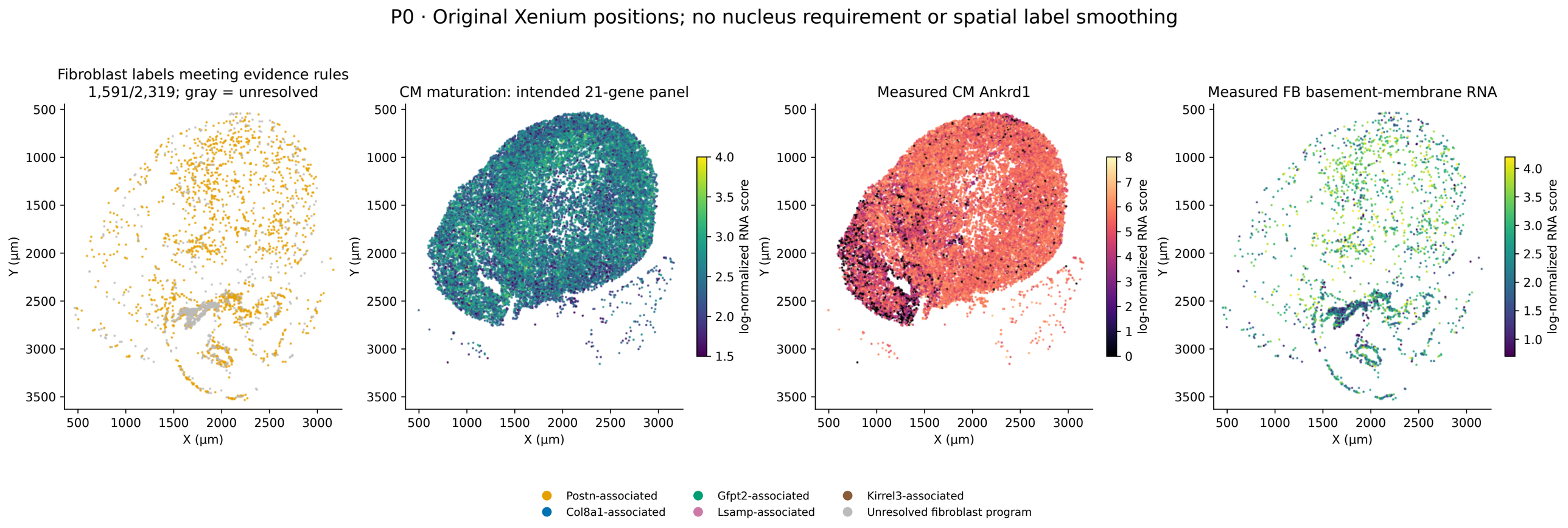
One spatial section per age. No lineage ancestry, matrix mechanics, or causal signaling is established.
Spatial-block intervals describe variation within the observed section, not independent animals.

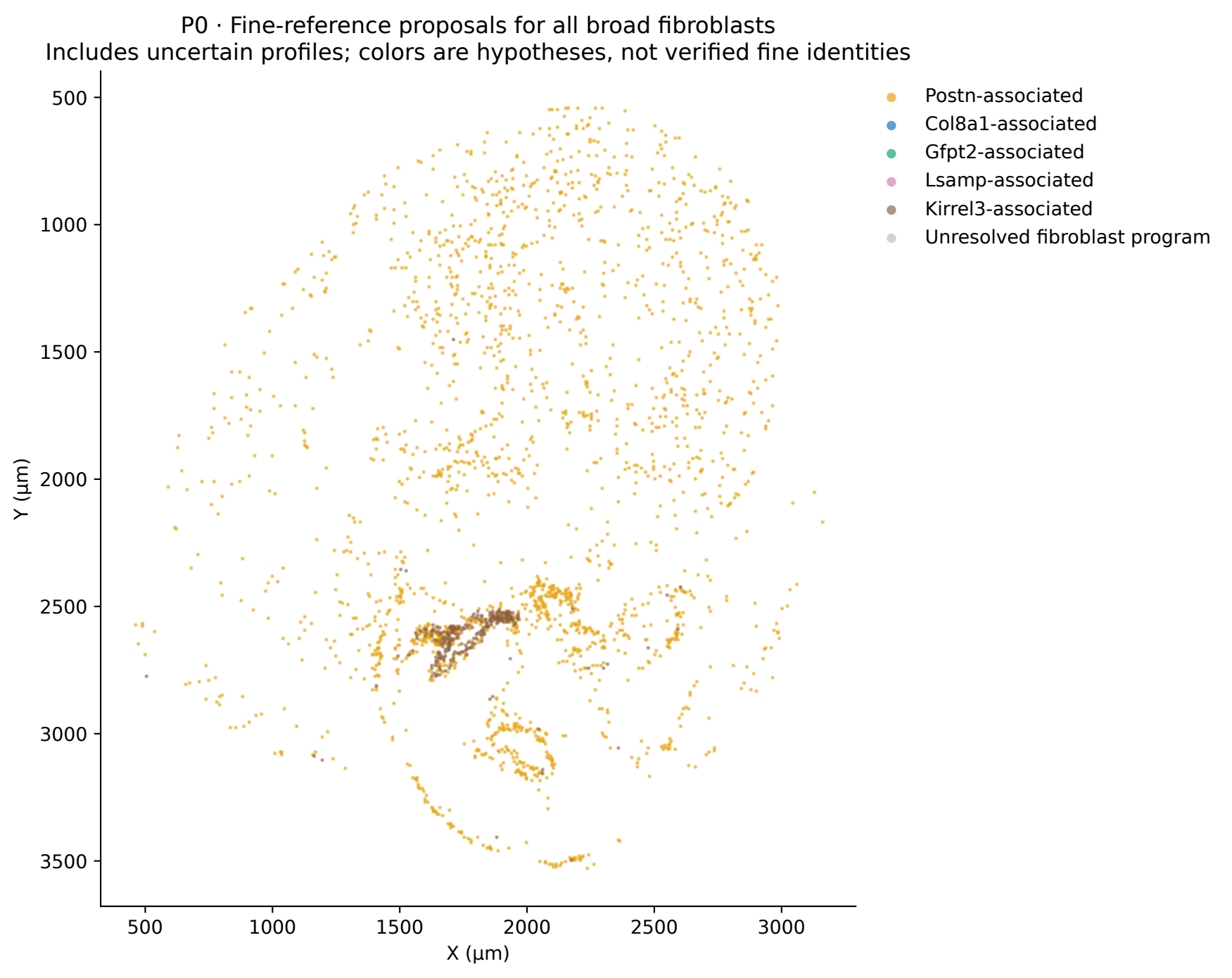

Fibroblast RNA groups within each stage
* residual lineage-mixing review; independent UMAP coordinates between stages
P0 · n=2,227
P7 · n=2,951
P14 · n=5,396
P21 · n=4,411
UMAP1
UMAP2
F0*
F4
F2
F6
F1
F5
F3
F0*
F4
F6
F2
F7
F1
F8
F3
F5*
F0
F7
F1
F4
F5
F2
F3
F6*
F3
F2
F6*
F5
F4
F1
F0
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

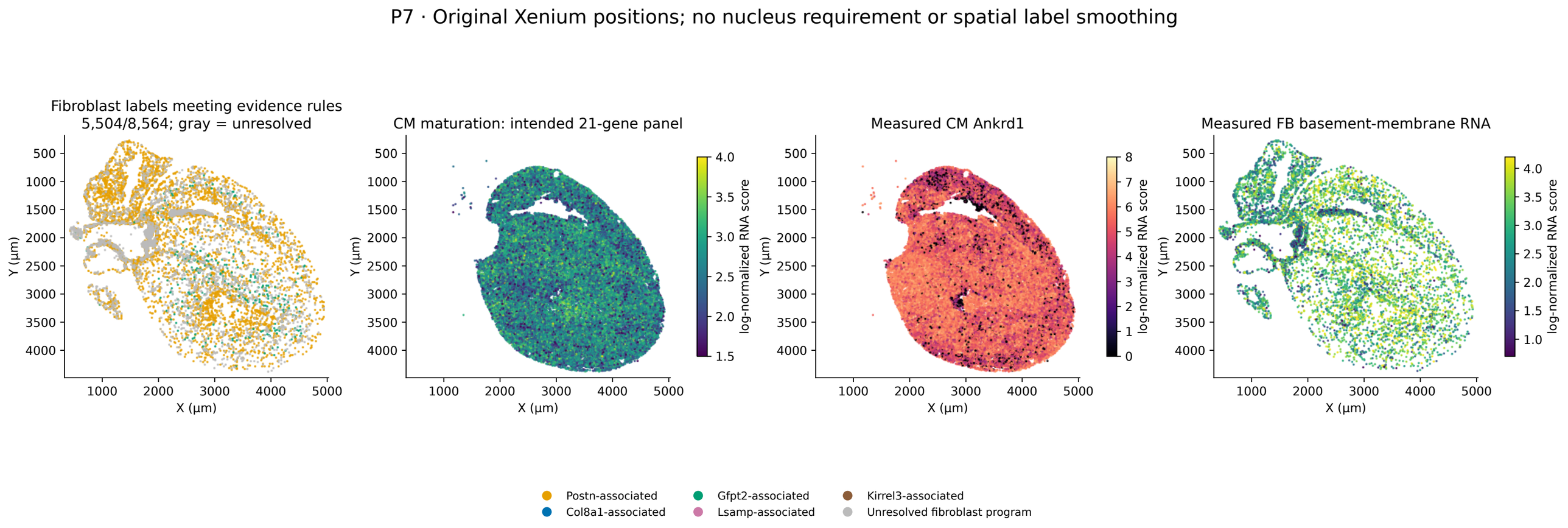

Stage-associated changes in the sampled fibroblast compartment
Counts reflect nucleus recovery and operational labels; they do not establish population expansion or ancestry
Fraction of reviewed RNA reference nuclei
1.0
0.8
0.6
0.4
0.2
0.0
P0
P7
P14
P21
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

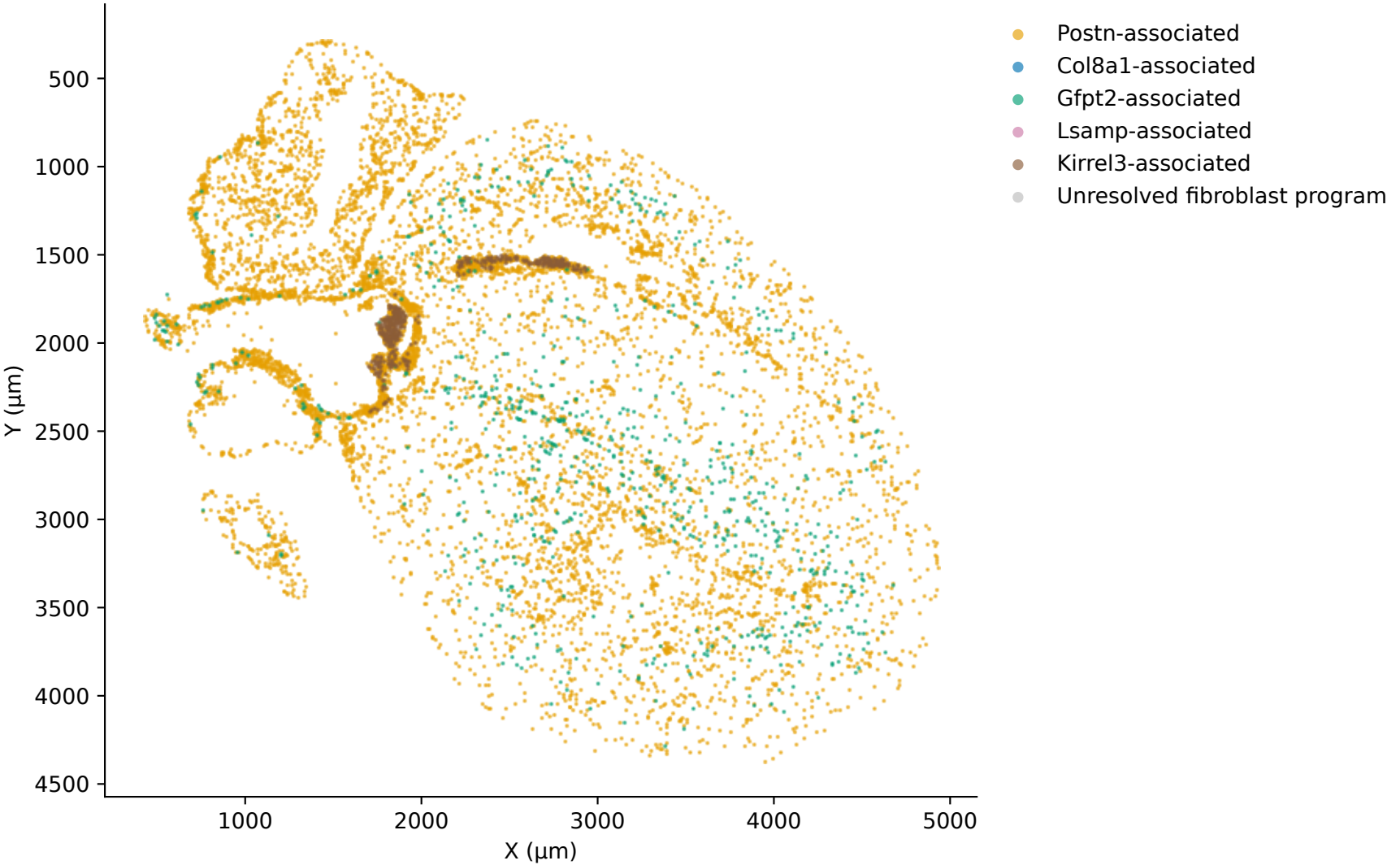

Matrix RNA varies with fibroblast program and age
Annotation genes withheld; two library pseudobulks averaged; each program standardized separately
P0 · Postn-associated
P0 · Kirrel3-associated
P7 · Postn-associated
P7 · Gfpt2-associated
P7 · Kirrel3-associated
P14 · Col8a1-associated
P14 · Gfpt2-associated
P14 · Lsamp-associated
P14 · Kirrel3-associated
P21 · Col8a1-associated
P21 · Gfpt2-associated
P21 · Lsamp-associated
P21 · Kirrel3-associated
P21 · Unresolved fibroblast program
Fibronectin
Fibrillar collagen
Basement membrane
Matricellular
Elastic fiber associated
Collagen processing crosslinking
Hyaluronan synthesis turnover
Relative program score (z)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0

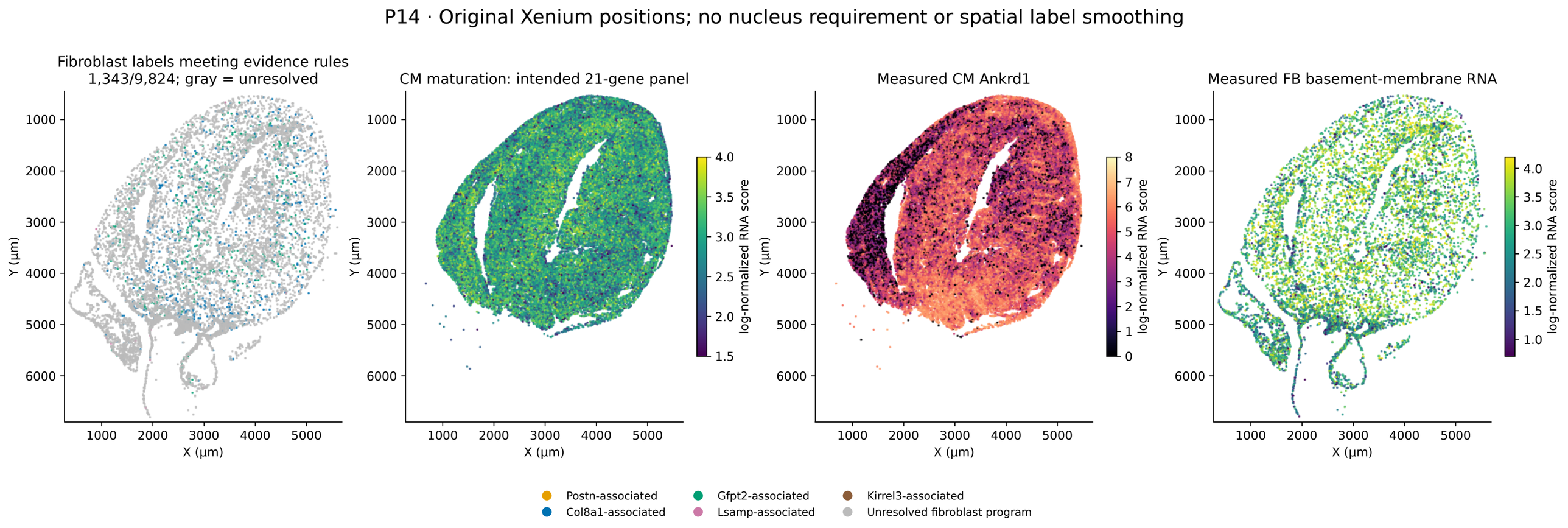

RNA resemblance with interpreted gene panels withheld
Same-stage background removed; correlation does not demonstrate ancestry
P0_F1
P0_F2
P0_F3
P0_F4
P0_F5
P0_F6
P7_F1
P7_F2
P7_F3
P7_F4
P7_F6
P7_F7
P7_F8
P14_F0
P14_F1
P14_F2
P14_F3
P14_F4
P14_F5
P14_F7
P21_F0
P21_F1
P21_F2
P21_F3
P21_F4
P21_F5
Pearson correlation
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

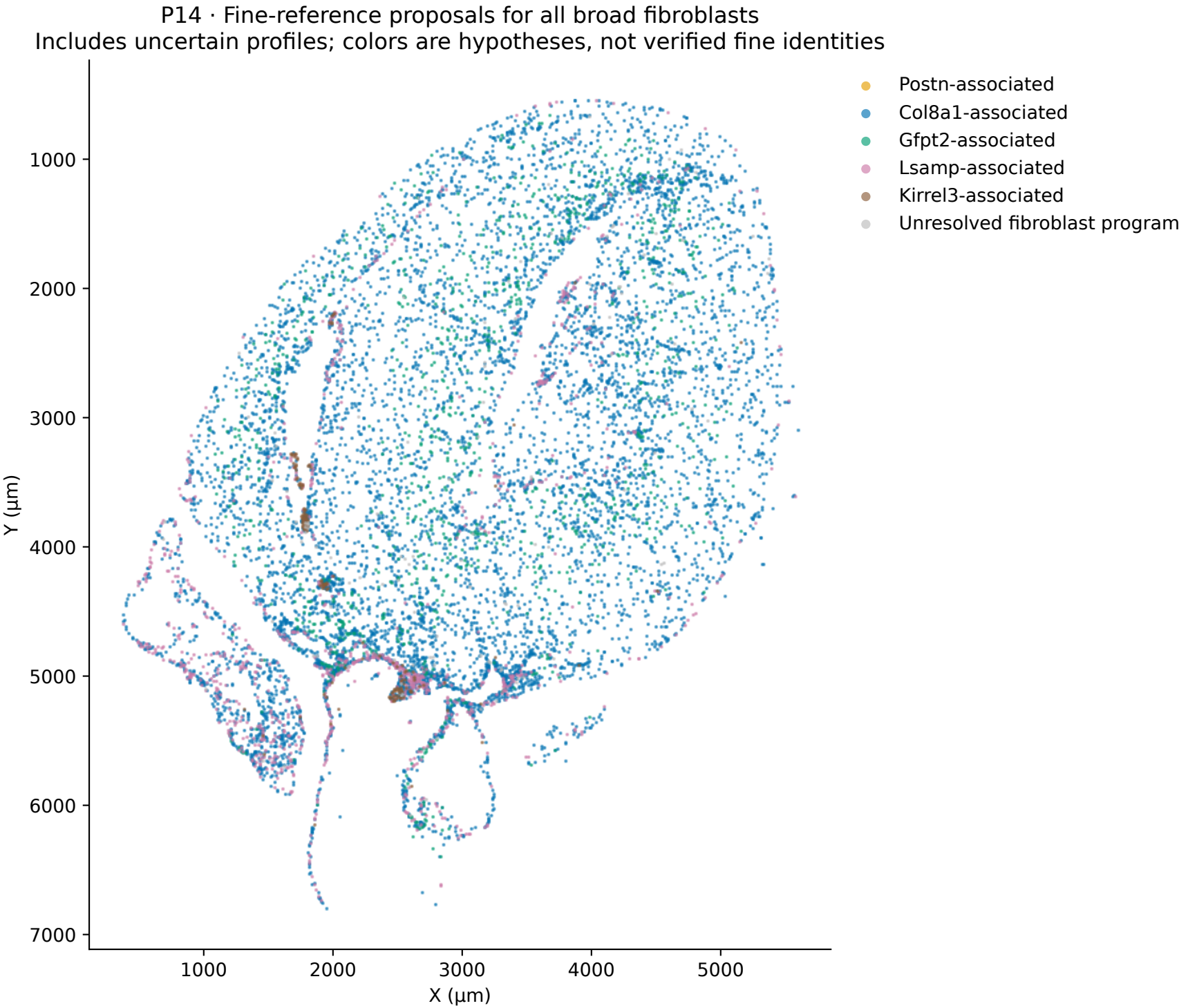

P0 · Original Xenium positions; no nucleus requirement or spatial label smoothing
Fibroblast labels meeting evidence rules
1,591/2,319; gray = unresolved
CM maturation: intended 21-gene panel
Measured CM Ankrd1
Measured FB basement-membrane RNA
Y (µm)
X (µm)
log-normalized RNA score
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

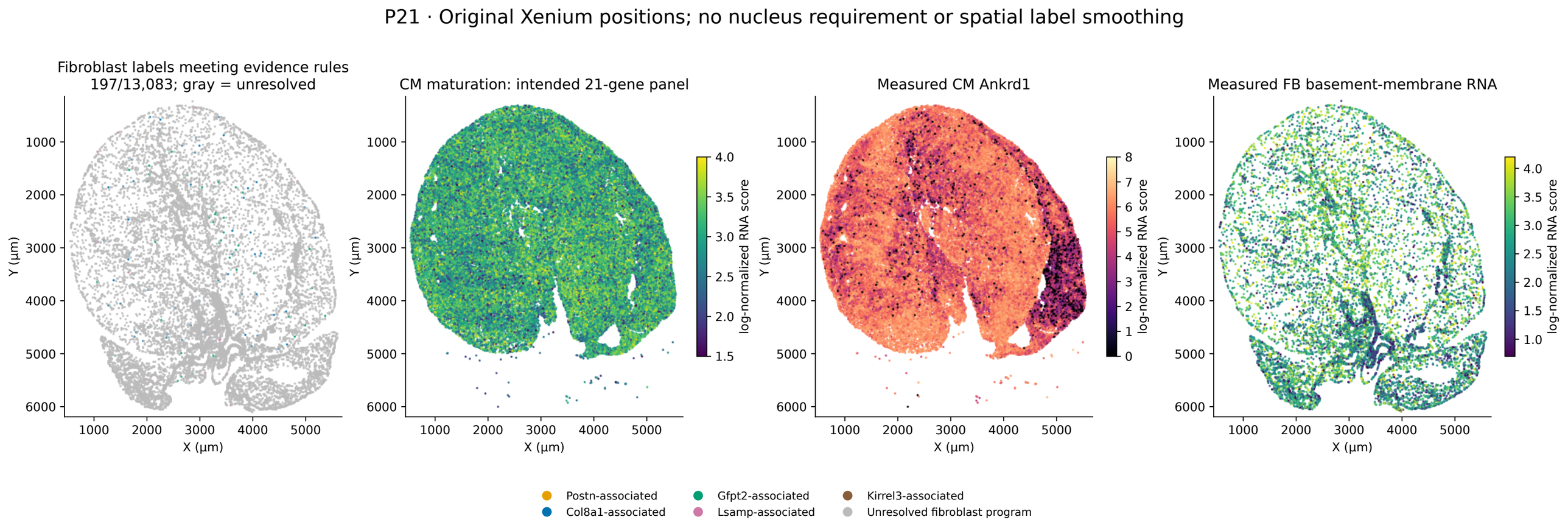

P0 · Fine-reference proposals for all broad fibroblasts
Includes uncertain profiles; colors are hypotheses, not verified fine identities
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program
Y (µm)
X (µm)
500
1000
1500
2000
2500
3000
3500

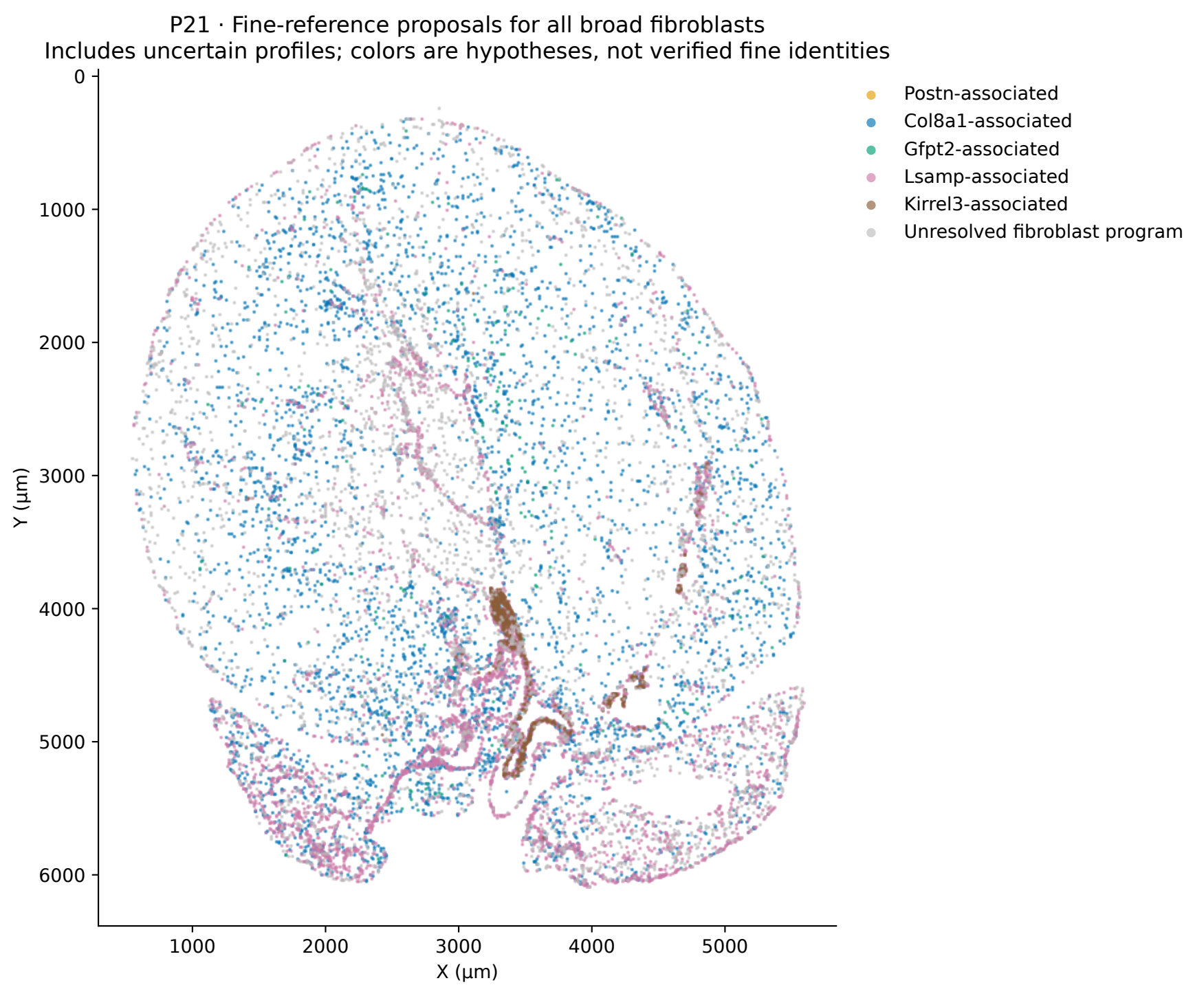

P7 · Original Xenium positions; no nucleus requirement or spatial label smoothing
Fibroblast labels meeting evidence rules
5,504/8,564; gray = unresolved
CM maturation: intended 21-gene panel
Measured CM Ankrd1
Measured FB basement-membrane RNA
Y (µm)
X (µm)
log-normalized RNA score
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

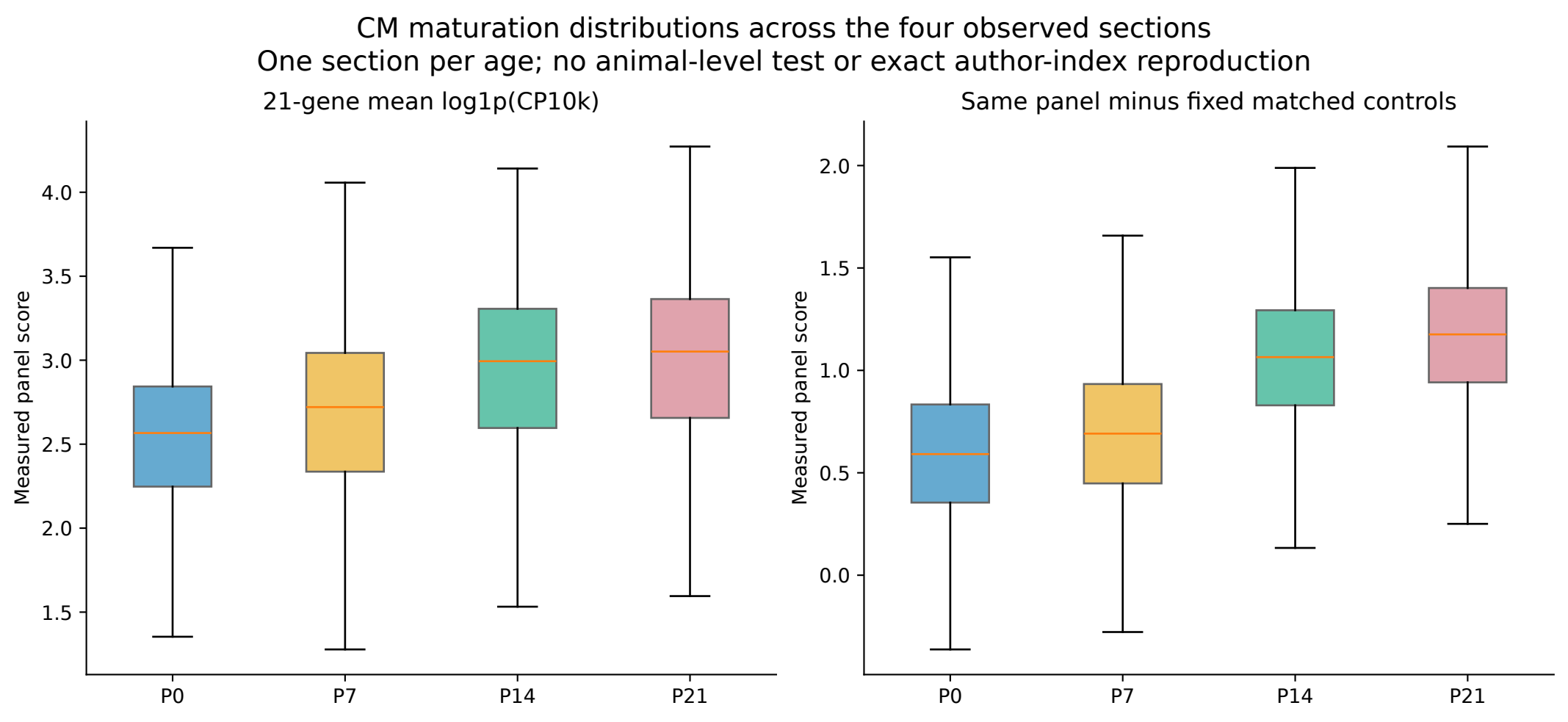

P7 · Fine-reference proposals for all broad fibroblasts
Includes uncertain profiles; colors are hypotheses, not verified fine identities
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program
Y (µm)
500
1000
1500
2000
2500
3000
3500
4000
4500
X (µm)
1000
2000
3000
4000
5000

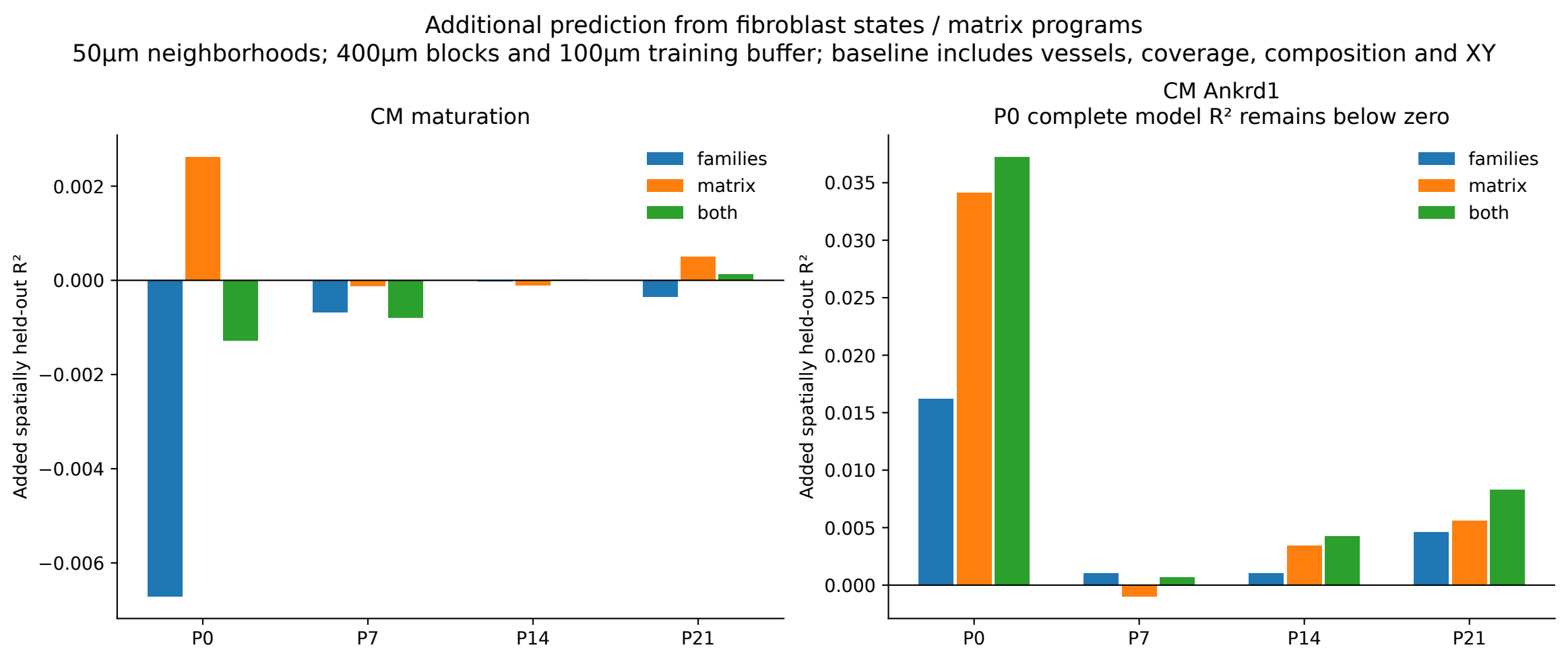

P14 · Original Xenium positions; no nucleus requirement or spatial label smoothing
Fibroblast labels meeting evidence rules
1,343/9,824; gray = unresolved
CM maturation: intended 21-gene panel
Measured CM Ankrd1
Measured FB basement-membrane RNA
Y (µm)
X (µm)
log-normalized RNA score
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

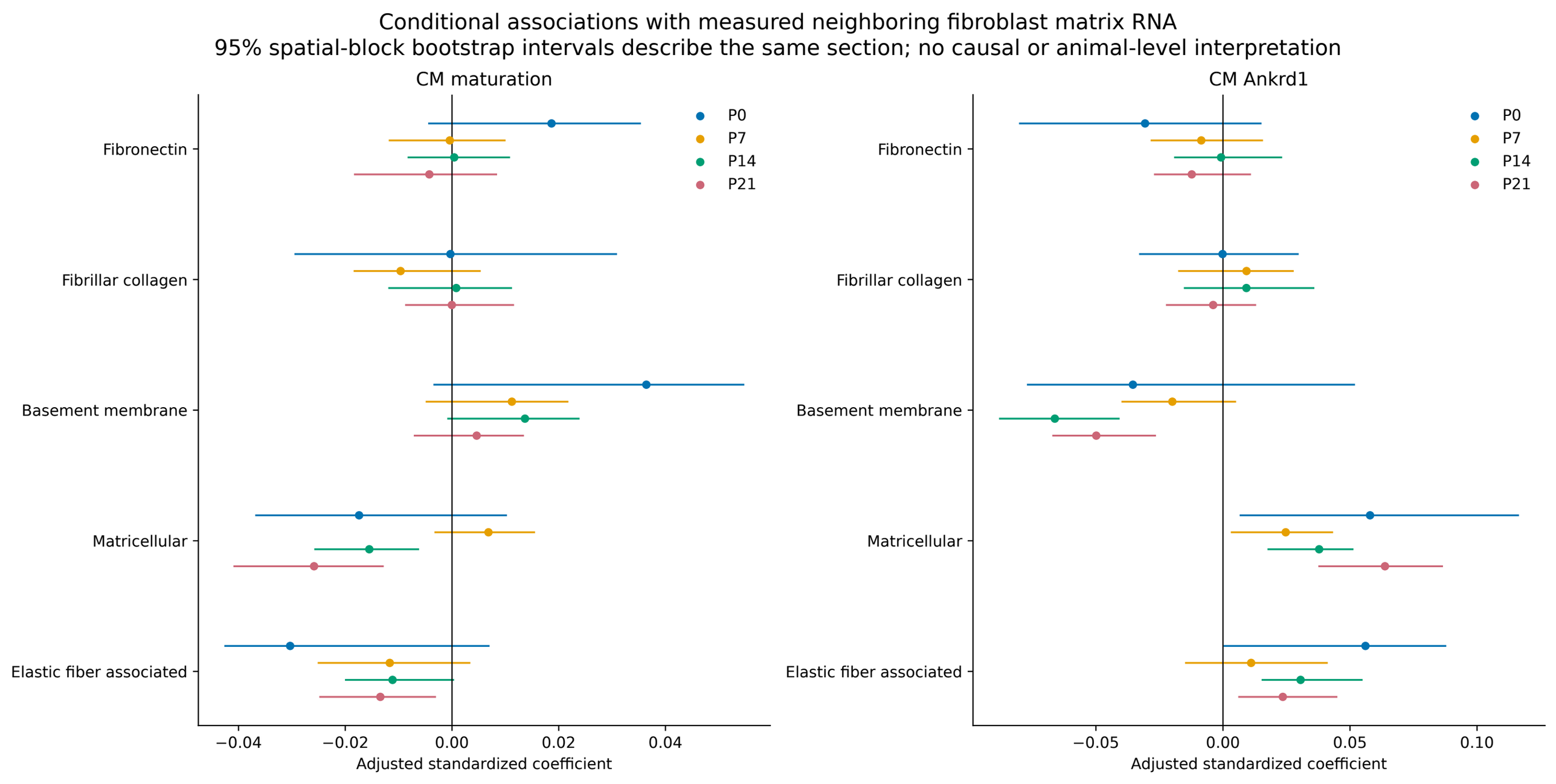

P14 · Fine-reference proposals for all broad fibroblasts
Includes uncertain profiles; colors are hypotheses, not verified fine identities
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program
Y (µm)
1000
2000
3000
4000
5000
6000
7000
X (µm)
1000
2000
3000
4000
5000

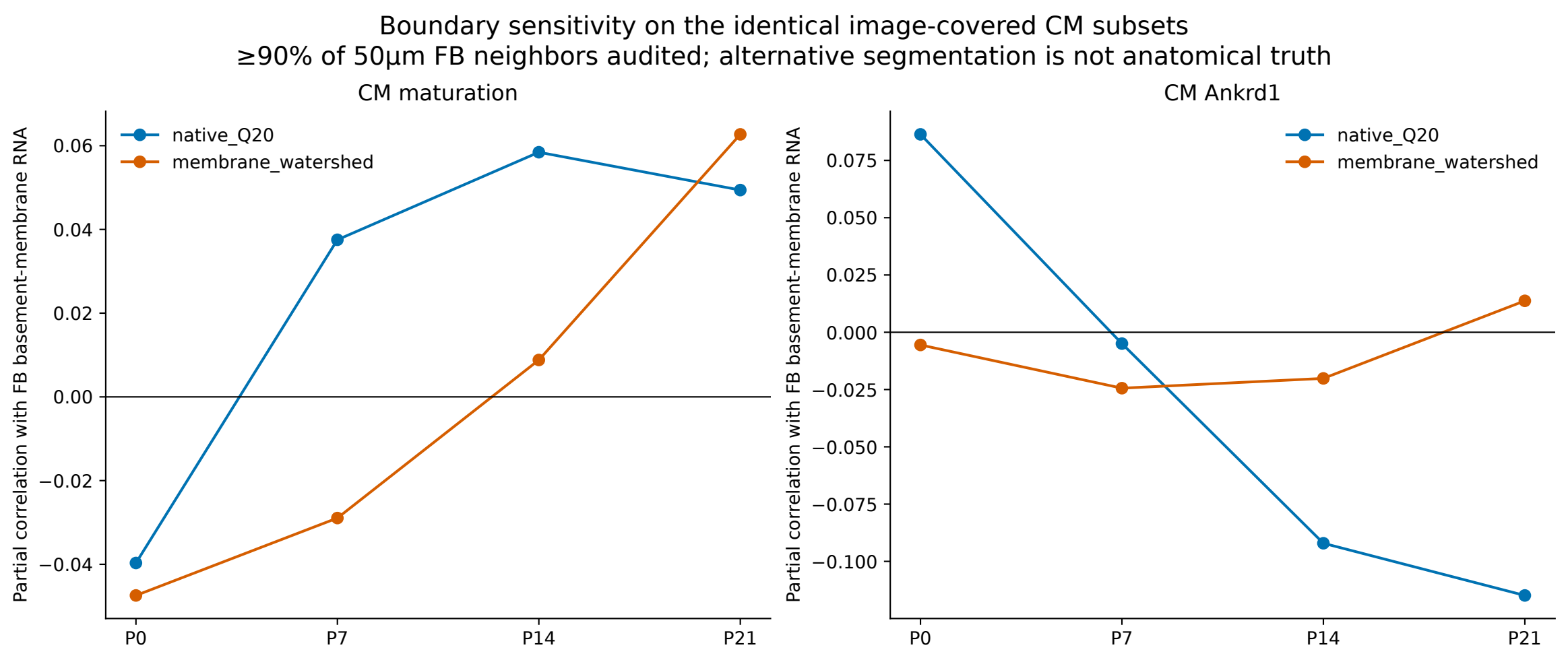

P21 · Original Xenium positions; no nucleus requirement or spatial label smoothing
Fibroblast labels meeting evidence rules
197/13,083; gray = unresolved
CM maturation: intended 21-gene panel
Measured CM Ankrd1
Measured FB basement-membrane RNA
Y (µm)
X (µm)
log-normalized RNA score
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

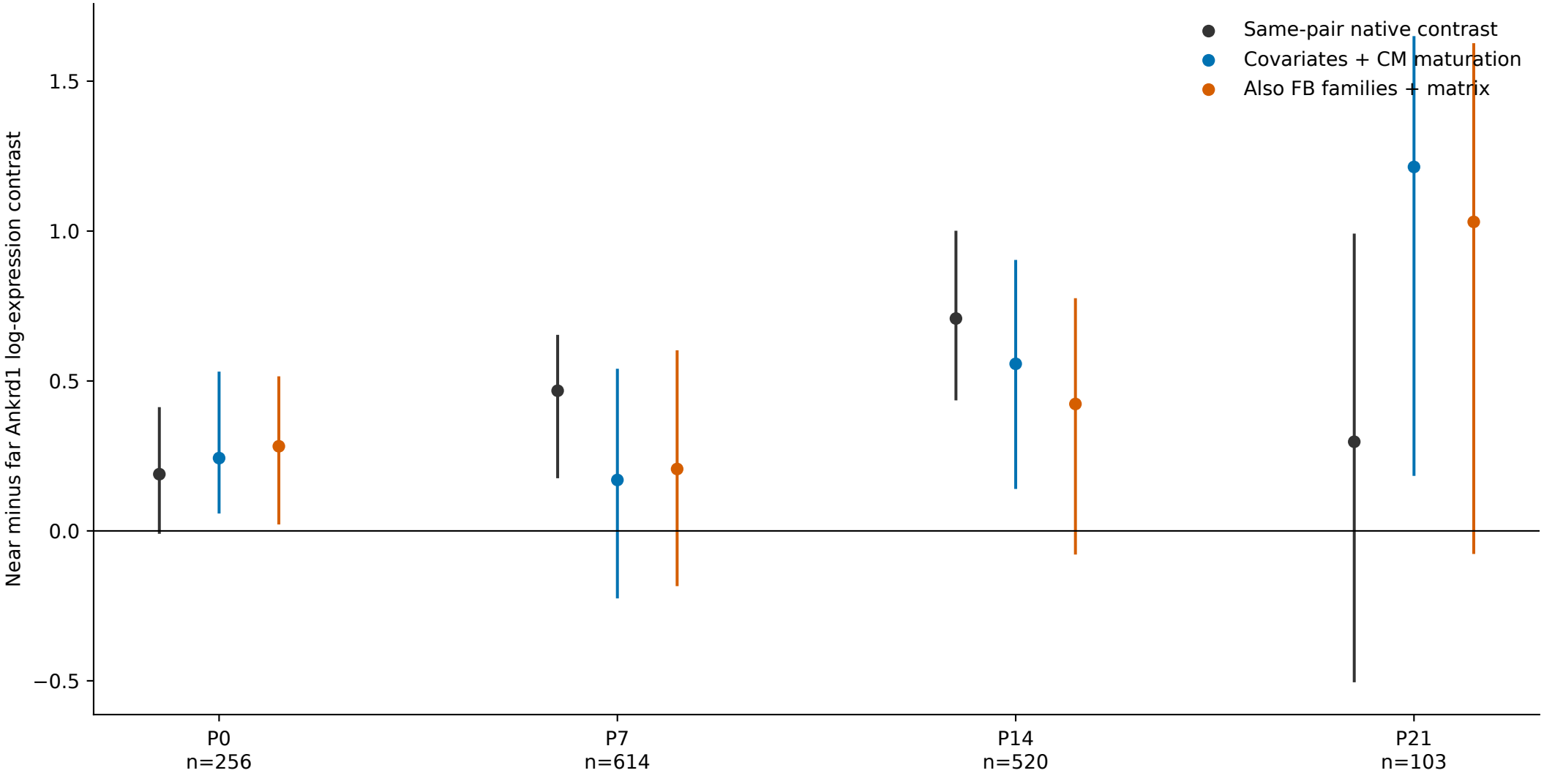

P21 · Fine-reference proposals for all broad fibroblasts
Includes uncertain profiles; colors are hypotheses, not verified fine identities
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000


CM maturation distributions across the four observed sections
One section per age; no animal-level test or exact author-index reproduction
21-gene mean log1p(CP10k)
Same panel minus fixed matched controls
Measured panel score
Measured panel score
4.0
3.5
3.0
2.5
2.0
1.5
2.0
1.5
1.0
0.5
0.0
P0
P7
P14
P21
P0
P7
P14
P21


Additional prediction from fibroblast states / matrix programs
50µm neighborhoods; 400µm blocks and 100µm training buffer; baseline includes vessels, coverage, composition and XY
CM maturation
families
matrix
both
Added spatially held-out R²
0.002
0.000
−0.002
−0.004
−0.006
P0
P7
P14
P21
CM Ankrd1
P0 complete model R² remains below zero
families
matrix
both
Added spatially held-out R²
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
P0
P7
P14
P21


Conditional associations with measured neighboring fibroblast matrix RNA
95% spatial-block bootstrap intervals describe the same section; no causal or animal-level interpretation
CM maturation
CM Ankrd1
P0
P7
P14
P21
Fibronectin
Fibrillar collagen
Basement membrane
Matricellular
Elastic fiber associated
−0.04
−0.02
0.00
0.02
0.04
−0.05
0.00
0.05
0.10
Adjusted standardized coefficient
Adjusted standardized coefficient


Boundary sensitivity on the identical image-covered CM subsets
≥90% of 50µm FB neighbors audited; alternative segmentation is not anatomical truth
CM maturation
CM Ankrd1
native_Q20
membrane_watershed
Partial correlation with FB basement-membrane RNA
0.06
0.04
0.02
0.00
−0.02
−0.04
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
P0
P7
P14
P21


Frozen vascular pairs after conditioning on fibroblast environments
Same neighborhood-supported subsets; segment-bootstrap intervals; P21 estimates unstable
Same-pair native contrast
Covariates + CM maturation
Also FB families + matrix
Near minus far Ankrd1 log-expression contrast
1.5
1.0
0.5
0.0
−0.5
P0
n=256
P7
n=614
P14
n=520
P21
n=103